# DOMINICK BRAICO

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## Education

**University of Michigan,** M.S. Robotics — Incoming August 2026

**University of Illinois Urbana-Champaign**, B.S. Mechanical Engineering, Minor in Computer Science — May 2026
Relevant Coursework: Software Development for Mobile Robots, Data Structures and Algorithms, Machine Learning, Robot Dynamics, Dynamics of Mechanical Systems, Signal Processing, Machine Design I/II, Electronic Circuits, CAD

## Work Experience

**Powertrain Engineering Intern** — May 2025 – Aug 2025
*Whisper Aero*
- Designed and released motor dynamometer test bench components in SolidWorks, creating formal engineering drawings with GD&T callouts (ASME Y14.5) and managing part revisions through SolidWorks PDM
- Instrumented dynamometer with thermocouple and current sensors for winding temperature monitoring and power characterization; conducted performance validation testing across the electric propulsor duty cycle
- Developed C/C++ firmware for FOC-based motor controllers and a Python GUI over CAN bus for real-time parameter tuning and telemetry

**Field Robotics Engineering Intern** — May 2024 – March 2025
*Robotics for Engineer Operations - Construction Engineering Research Laboratory*
- Designed a rugged electronic control module for robot sensors and compute unit, reducing electrical system volume and improving weatherproofing on a modular cable-driven construction robot
- Modeled and fabricated 10+ custom sensor mounts in SolidWorks for LiDAR and IMU systems, improving sensor data integrity on autonomous heavy equipment platforms
- Engineered a pulley system routing actuation forces from onboard motors to the central payload on Co3MaNDR, a cable-driven parallel robot for cooperative modular manipulation
- Validated software and hardware integration across mechanical, electrical, and autonomy subsystems, diagnosing cross-system failures and translating between sensor data formats and control inputs

## Extracurriculars & Leadership

**Control Systems & Sensor Integration Lead** — February 2024 - Present
*GHOST Electric Motorcycles*
- Configured CAN bus network for communication between motor controller, BMS, and onboard sensors; performed bench testing to validate electrical system performance prior to vehicle integration
- Implemented feedback control algorithms to tune torque response of a 45kW PMAC motor

## Projects

**Autonomous Moon Rover Excavation System**
- Fabricated and designed regolith collection mechanism transporting BP-1 lunar regolith simulant on a robotic excavation system competing in a NASA Artemis Challenge

**Robotic Car Heptathlon**
- Modeled transmission and chassis in Fusion 360, performing FEA on gears and shafts to validate durability under load with minimized cost and mass
- Optimized chassis as a single-print structure applying DFM principles, fabricating via 3D printing and laser cutting; delivered functional prototype under 20% of budget through iterative design refinement

**ASNE PEP Autonomous Surface Vessel Competition**
- Designed and fabricated mechanical mounting systems for onboard electronics, ESCs, and propulsion hardware, ensuring structural rigidity and water resistance under open-water operating conditions
- Ran ANSYS Fluent CFD simulations to characterize hull drag and FEA to analyze resulting structural stresses, validating propulsion system sizing against simulation-derived performance targets and structural support integration

## Skills

**Mechanical Design:** SolidWorks (Certified Mechanical Design Associate), Autodesk Fusion, GD&T, DFM
**Fabrication:** 3D Printing, Waterjet, Laser Cutter, Shop Tools, Soldering
**Programming:** Python, C/C++, ROS 2